

Sigvard Bratlie

Data Scientist



M.Sc. student in Data Science at NMBU with 4 years of project management experience in commercial real estate. Currently researching agentic AI for the legal sector.

Timeline

Data Science | NMBU

Aug 2024 - Present

- Researching agentic legal AI at Bahr AS for my master thesis — building an LLM-powered agent for case summarization, structured fact extraction, RAG on Norwegian law, and real-time chat assistance. Developing compression techniques for large-scale legal document processing and long-term context handling.
- Built a YOLO11 object detection model for Rumex identification in drone imagery. Also worked with CNNs, canny edge detection, and classical image processing techniques.
- Built an interactive Streamlit dashboard analyzing Norwegian electricity production/consumption vs. historical weather data, with time-series decomposition, forecasting, and outlier detection.
- Developed an end-to-end ETL pipeline scraping Finn.no, enriching with public registry data (Geonorge, Statens Vegvesen, Kartverket), and training pricing models with XGBoost/CatBoost/LightGBM on GCP.
- Built SIBR Market — an AI valuation platform where users chat with LLM agents for real estate and vehicle valuations, deployed on Google Cloud Run.

- Built an oil spill diffusion simulator with mesh-grid modeling and auto-generated MP4 visualizations.
- Coursework in statistical theory (Bayesian/frequentist, MLE, multivariate distributions), optimization (gradient descent, Newton's method, KKT, simplex), deep learning, and algorithm design.

Project Manager | Newsec

Aug 2020 - Jul 2024

- Managed multi-million NOK construction and tenant fit-out projects end-to-end, coordinating 20-30 stakeholders across consulting engineers, contractors, architects, clients, and municipal authorities. Portfolio included residential conversions, commercial rehabilitations, and public space developments across central Oslo.
- Delivered projects for institutional property owners and international investors — including OBOS, Pareto Securities, Clarkson Platou, SBB Norden, and Selvaag — handling both new developments and commercial tenant adaptations.
- Led technical due diligence on property transactions across office, retail, industrial, and residential assets, producing integrated assessments for both buy-side and sell-side clients.
- Managed full procurement cycles — tender processes, bid evaluations, contract negotiation, and administration under NS8407/NS8402 frameworks — including public sector procedures and regulatory applications.
- Owned project financial reporting and ensured compliance with regulatory frameworks and sustainability standards including BREEAM.

Engineering | NTNU

Aug 2017 - Jun 2020

- B.Sc. in Structural Engineering with focus on construction management, structural design, and regulatory frameworks.
- Completed a feasibility study for the redevelopment of Mustad Industrial Park as part of a real estate development course.
- Bachelor thesis on replacing structural steel systems with timber — analyzing environmental impact, cost implications, and structural performance.

- Foundation in project planning, contract management, and navigating building regulations and public approval processes.

Additional Experience

- Independently rezoned and repositioned a mixed-use property in Hamar from office/residential to 10 residential units, managing the full regulatory process before selling the project with building permits to Odin Gruppen AS.
- Continued taking on advisory and project management assignments from former clients on an hourly basis alongside studies.
- Economics coursework at BI Norwegian Business School (2021-2022).

Skills

Machine Learning

- ANN (PyTorch & TensorFlow)
- Transformers (HuggingFace)
- ML (Scikit Learn, XGBoost, ++)
- Agents (Langchain/langgraph)
- Data handling (Pandas, NumPy, Spark)
- Plotting (Matplotlib, Seaborn, Plotly)
- Computer vision (OpenCV, YOLO)

Cloud Computing

- GCP, Azure & AWS
- Data warehouse (Big Query, Snowflake)
- CI/CD tools (Cloud Build, etc.)
- Serverless computing (Cloud Run)
- Docker
- VM and Kubernetes
- NoSQL DB (MongoDB, Firestore)
- ++ (Secrets, IAM handling, etc.)

Other Tools/Skills

- FastAPI
- Git
- Building SDK
- Streamlit
- Dashboards (PowerBI, Looker)
- Figma

Professional Skills

- Project Management
- Client Relations
- Business Acumen
- Reporting & Presentation

Projects

- [Agentic Legal AI \(Master Thesis\)](#): LLM/NLP project for legal document processing.
- [Weather & Electricity Dashboard](#): Data analysis and EDA dashboard.
- [Rumex Detection Model](#): Computer vision model using YOLO.
- [Market Data ETL Project](#): Data fetching and scraping pipeline.
- [SIBR Market](#): UI/LLM-based market interface.

Contact Info

Phone: +47 938 49 732

[LINKEDIN](#)

[GITHUB](#)

[EMAIL](#)